

Nayak adds, "Aspirants must be strong in power semiconductors application engineering in power system, as there are tremendous opportunities and also continuous improvement in design of semiconductors focusing on cost and materials used."

Participating in industrial seminars, conferences and exhibitions related to power electronics systems and products would help improve one's knowledge base.

Agarwal says, "An internship in the power electronics division of reputed organisations would help one gain a good experience of designing and fabricating at least some basic power electronics systems."

Short-term courses to fine-tune skills

When it comes to short-term courses, there are just a few of them. Some institutes offer advanced training in power electronics for graduates, while others offer courses in power devices, motors and motor controls, AC drives, etc. A few institutes also offer certificate courses in power electronics, power supply design and switched-mode supplies.

Garadi shares, "I do not see any short-term course in power electronics doing full justice to the topic. However, there are a lot of master's programmes in power electronics offered by many premier technical institutes. IITs, these days, are offering integrated B.Tech and M.Tech programmes in power electronics."

Nayak concedes, "Best learning is achieved through on-the-job training at the industries as they give tremendous opportunities and challenges for harnessing the skills and also application of their basic knowledge leading to product design and solutions."

Sharing with us how Bosch spends time and effort in training and sharpening skills of new recruits, Garadi says, "Freshers go through intensive

"In addition to the industrial sector, power electronics is rapidly gaining momentum in automotive space too with the advent of electric and hybrid vehicles. Power electronics engineers command a better pay package than mixed-signal board designers. The exact pay package depends on the company recruiting plus the specific skills that the company is seeking" — Basavaraj Garadi, chief expert, Robert Bosch Engineering and Business Solutions

training both in generic topics that they are likely to be associated with and specific domain that they get into."

He adds, "We are continuously improving the training programmes and the content. We are bringing in hands-on training with a few sample design tasks so that the engineers realise and appreciate the challenges and get an idea of the serious nature of engineering."

Times are changing

Industry experts believe that power electronics is an emerging field in India. In spite of being an interesting core engineering area of interdisciplinary nature with a lot of challenges, it is believed to be less popular.

Agarwal feels, "This is probably because of a general bias against the field of 'power,' which is considered to be an 'old' and well established field."

He further adds, "Until a few decades ago, a power electronics engineer could find a job only in homegrown private companies (which were very few) and select government establishments such as BHEL, DRDO and ISRO. But with the coming of multinational companies (MNCs) and the increasing thrust on renewable energy conditioning, automotive industry, efficient lighting systems and electric drives, available opportunities and salaries have increased significantly."

Garadi says, "All of these sectors offer numerous employment opportunities to graduates as well as postgraduates. There are both large

MNCs as well as SMEs that pick professionals in power electronics on a regular basis. Incidentally, last year, our own company recruited a few from premier IITs."

"With power electronics visible in almost all spheres of life, and with increasing competition among the local industries and the growing need for power electronics, a sharp rise in the demand for power electronics experts is inevitable," believes Agarwal.

In the future

The global need for core engineering is increasing, which is one of the reasons why many of the global companies are setting up research and engineering units outside their base.

Nayak says, "Core engineering is very much in demand as there will be a huge capacity build up with country's plan of 100 GW during the current Planning period, i.e., 2012-17."

He adds, "Considering recent developments in the field of power systems, all-round focus on renewable energy and drive towards efficiency and bulk transmission, both in India and abroad, there is enough scope for power electronics. There will be tremendous placement opportunities for those who are willing to pursue a career in power electronics."

IET Power panel, headed by Nayak, consists of representatives from industries, academia and research units, focuses on ways to develop the core engineering resources and how to make them employable through the panel's young professional and student chapter activities with continuous interaction and guidance. ●

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"There is a scarcity of expert manpower in this field" — Vivek Agarwal, professor, Department of Electrical Engineering, Indian Institute of Technology-Bombay